



Basic Principles of Medicine 2
DM: Digestive System & Metabolism | Lecture 7

Anatomy of the Large Intestine

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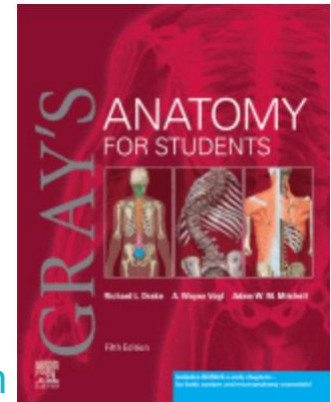
Objectives

Midgut & Hindgut Structures

- | | |
|--------------------------------|--|
| SOM.MK.1.BMP2.3.DM.1.ANAT.1140 | Explain the passage tumor cells may take via the lymphatic drainage from different portions of the small and large intestine. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1142 | Explain which parts of the GI tract may become involved in volvulus. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1146 | Explain the surgical approach through the layers of the abdominal wall to approach an inflamed appendix and the surgical significance of the muscle fiber orientation |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1148 | Describe the location of the large intestine and its relationship to other organs. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1149 | Describe the different portions of the large intestine and explain the external and internal differences between these parts. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1150 | Describe the blood supply and the venous and lymphatic drainage of the large intestine. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1151 | Describe the location, arterial supply, and the venous and lymphatic drainage of the rectum and anal canal. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1152 | Describe the location of the pectinate line and the changes that occur at the pectinate line including arterial supply, venous and lymphatic drainage, innervation and embryological origin. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1153 | Describe the anastomotic communications between vessels of the foregut and the midgut and between midgut and hindgut. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1154 | Explain how the hindgut receives its blood supply after repair of an abdominal aortic aneurysm since the inferior mesenteric artery is usually sacrificed during the procedure |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1155 | Explain why surgical resection of the ascending or descending colon should be done in the lateral paracolic gutters. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1156 | Describe the structural modifications and clinical signs of diverticulosis. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1157 | Differentiate between small and large bowel obstruction and their clinical presentation. |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1158 | <i>Define enterocele and rectocele (independent student review)</i> |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1181 | Describe the portal circulation and the areas of anastomosis with the systemic circulation. |

Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Online Textbook: [Chapter 4: Click here](#)
 - Abdominal Viscera
 - Large Intestine
 - Cecum and Appendix, Colon
 - Arterial Supply, Venous Drainage, Lymphatics
 - In the Clinic: Diverticular disease, Bowel obstruction
- Pay special attention to the **green (print text)/blue (ebook)** “In the Clinic” boxes for the clinical correlations for this topic



Midgut

Extends from the 2nd part Duodenum distal to the major duodenal papilla to beginning the distal 1/3 of the transverse colon

Includes:

- Jejunum, Ileum (small intestines)
- Cecum, Appendix
- Ascending colon
- Proximal 2/3 of the transverse colon

Arterial supply: Superior Mesenteric a.

Venous drainage: Superior mesenteric v.

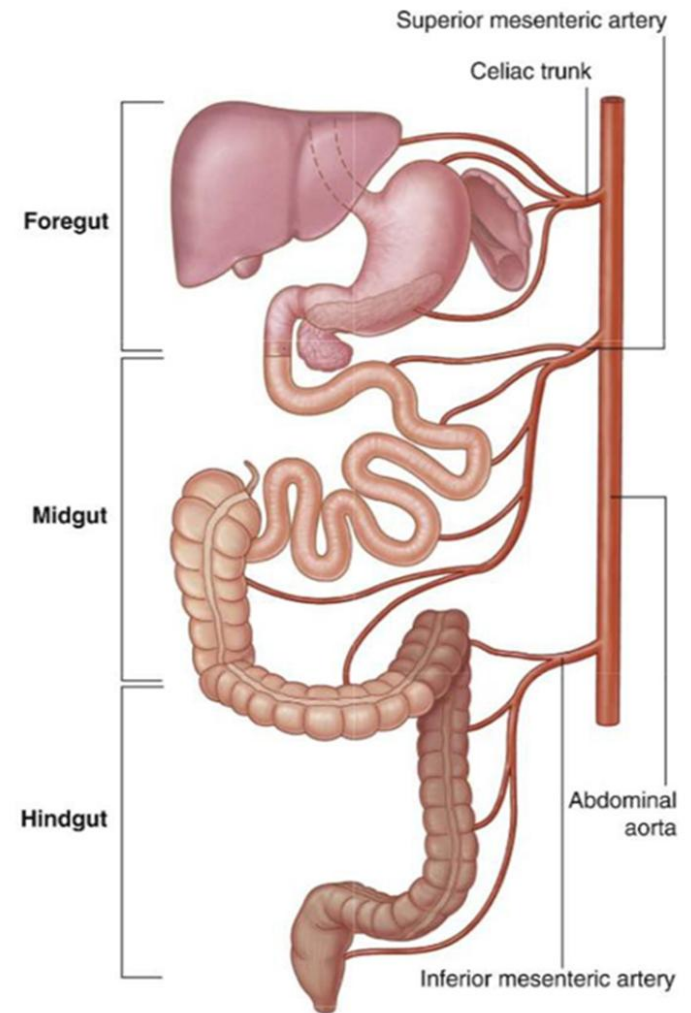
Lymphatic drainage: Superior Mesenteric nodes

Nerve supply:

Parasympathetics: Vagus nerve (*preganglionic*)

Sympathetics: Lesser (T10-T11) and Least (T12) thoracic splanchnics (*postganglionic*)

Visceral afferents: travel along sympathetic nerve fibers T10-T11



Hindgut

Begins at the junction of the proximal 2/3 and distal 1/3 transverse colon and ends at the pectinate line.

includes

- Distal 1/3 transverse colon
- Descending colon
- Sigmoid colon
- Rectum
- Superior/Upper 2/3 of Anal Canal

Arterial supply: Inferior Mesenteric a.

Venous drainage: Inferior mesenteric v.

Lymphatic drainage: Inferior Mesenteric nodes

Nerve supply:

Parasympathetic: Pelvic splanchnics S2,3,4 (preganglionic)

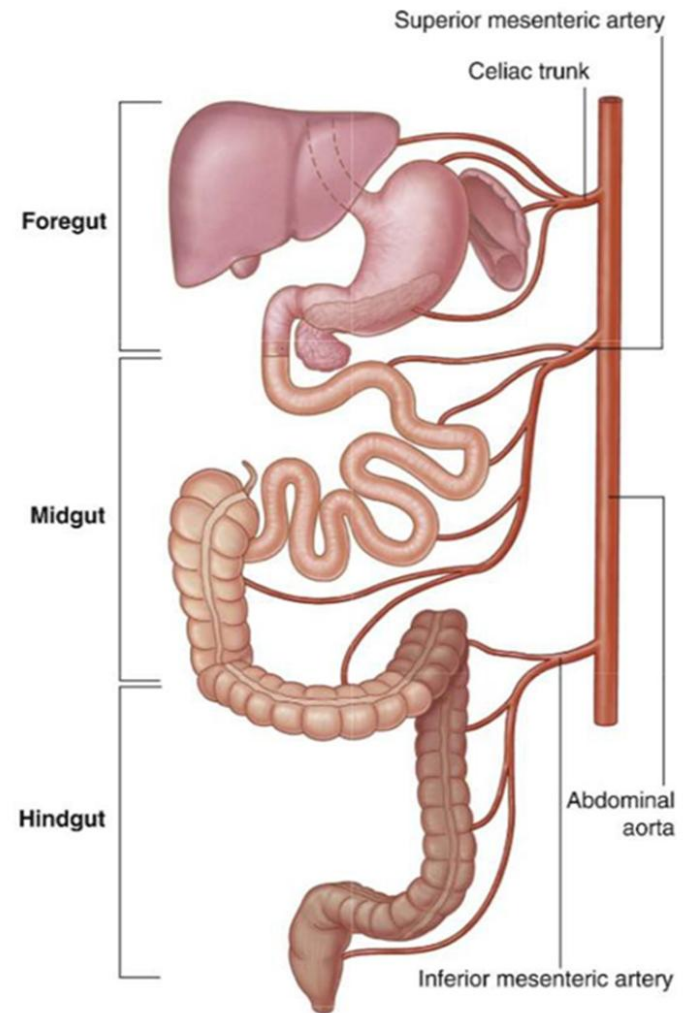
Sympathetic: Lumbar splanchnics L1-L2 (postganglionic)

Visceral afferents:

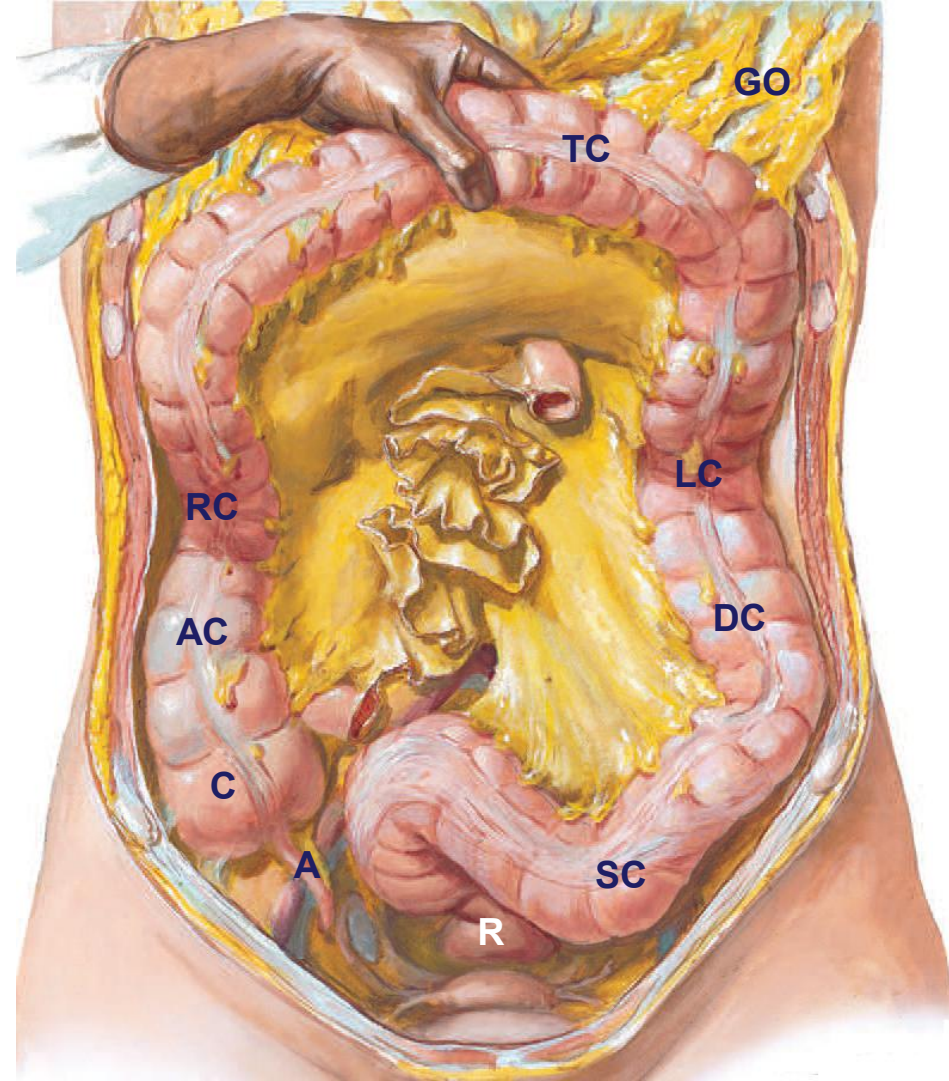
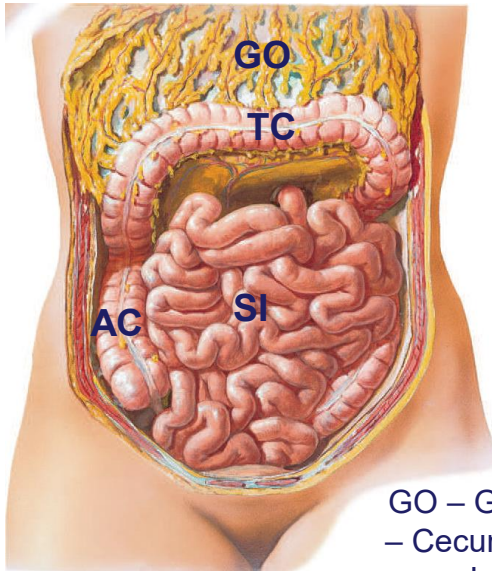
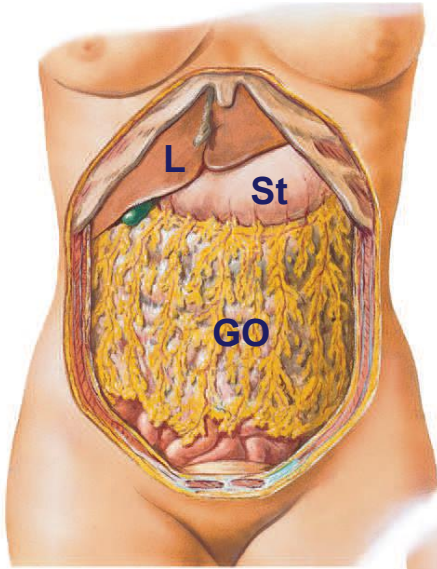
Above pelvic pain line: travel along Sympathetic nerve fibers: T12-L2

Below pelvic pain line: travel along parasympathetic nerve fibers: pelvic splanchnics S2,3,4

NOTE: PPL runs through the midpoint of the sigmoid colon



Large Intestine



GO – Greater omentum; L – Liver; St – Stomach; SI – Small intestines; A – Appendix; C – Cecum; AC – Ascending colon; RC – Right colic Flexure; TC – Transverse colon; LC – Left colic Flexure; DC – Descending colon; SC – Sigmoid colon; R - Rectum

Large Intestine

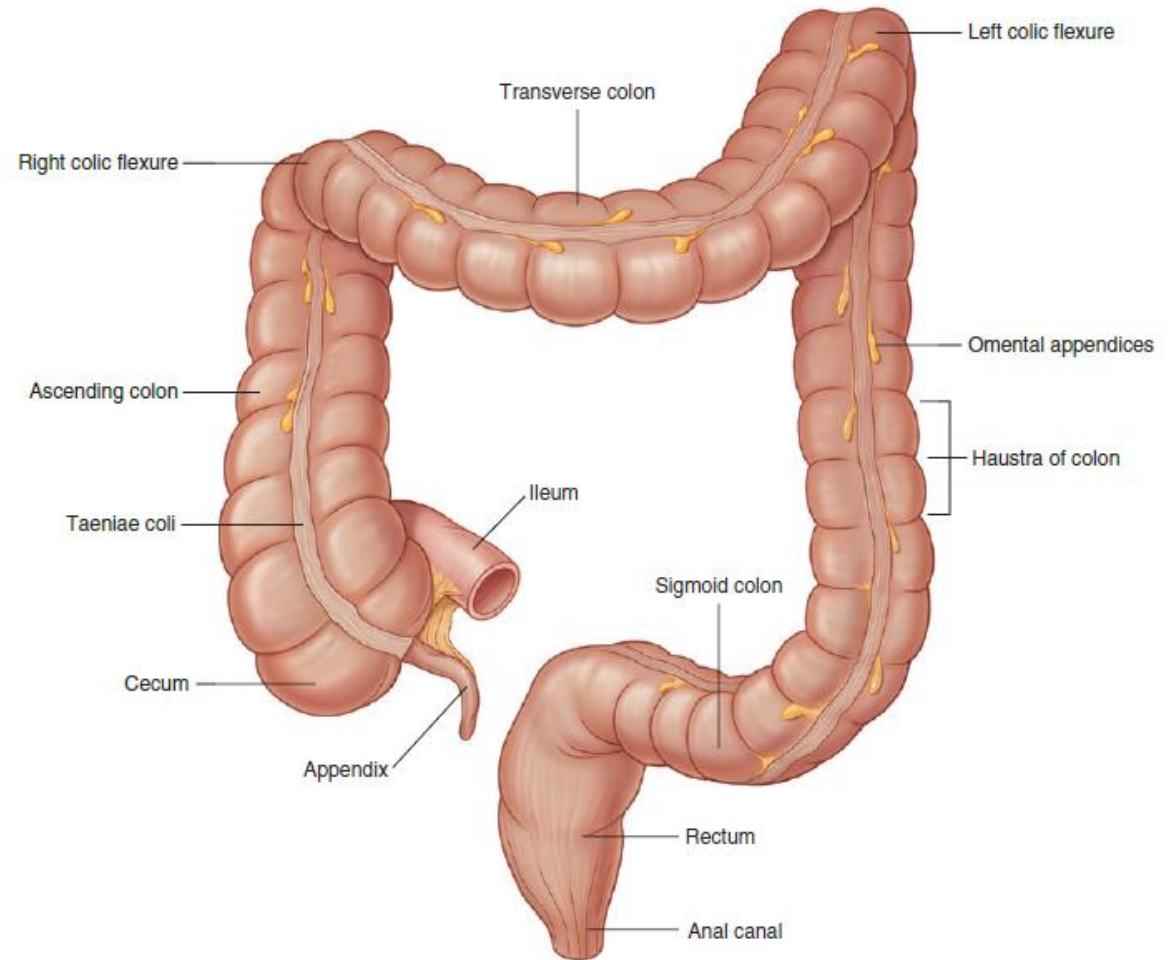


Fig. 4.79 Large intestine.

- Distal ileum to anus
- Diameter larger than small intestine

Parts of Large Intestine

- **Cecum**

- Broad, blind intestinal pouch, inferior to ileocecal valve in Right iliac fossa,
- Continuous with ascending colon; Mobile
 - Anterior + posterior cecal (Ileocolic a.)

- **Appendix**

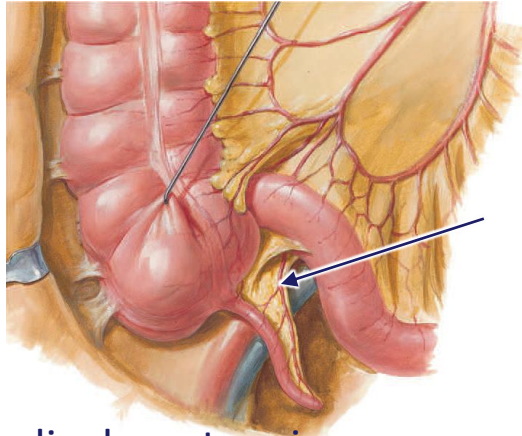
- Narrow, hollow, blind-ended tube connected to cecum (posteromedially)
 - Entry into cecum consistent – taeniae coli can be followed to base of appendix
- Large lymph aggregates; mesoappendix with appendicular vessels
 - Appendicular (Ileocolic a.)

- **Colon**

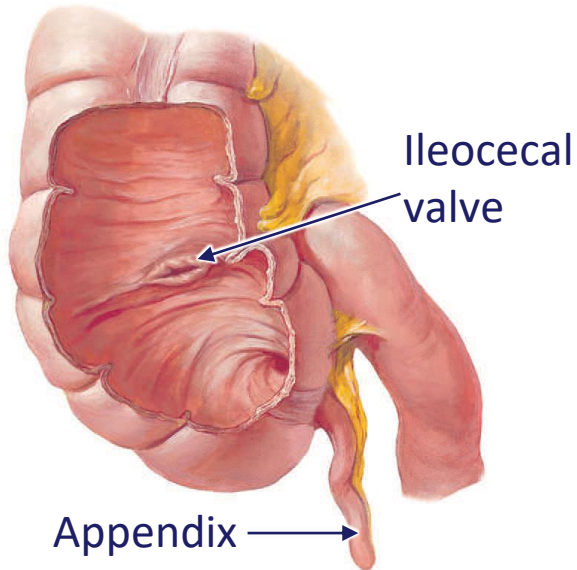
- Colic flexures, Paracolic gutters, Marginal artery
- Ascending – colic branches of ileocecal a., right colic a (SMA)
- **Transverse** – right colic + middle colic (SMA); left colic a. (IMA)
- Descending – left colic a. (IMA)
- Sigmoid – sigmoidal arteries (IMA)

- **Rectum & Anal Canal**

Large Intestine

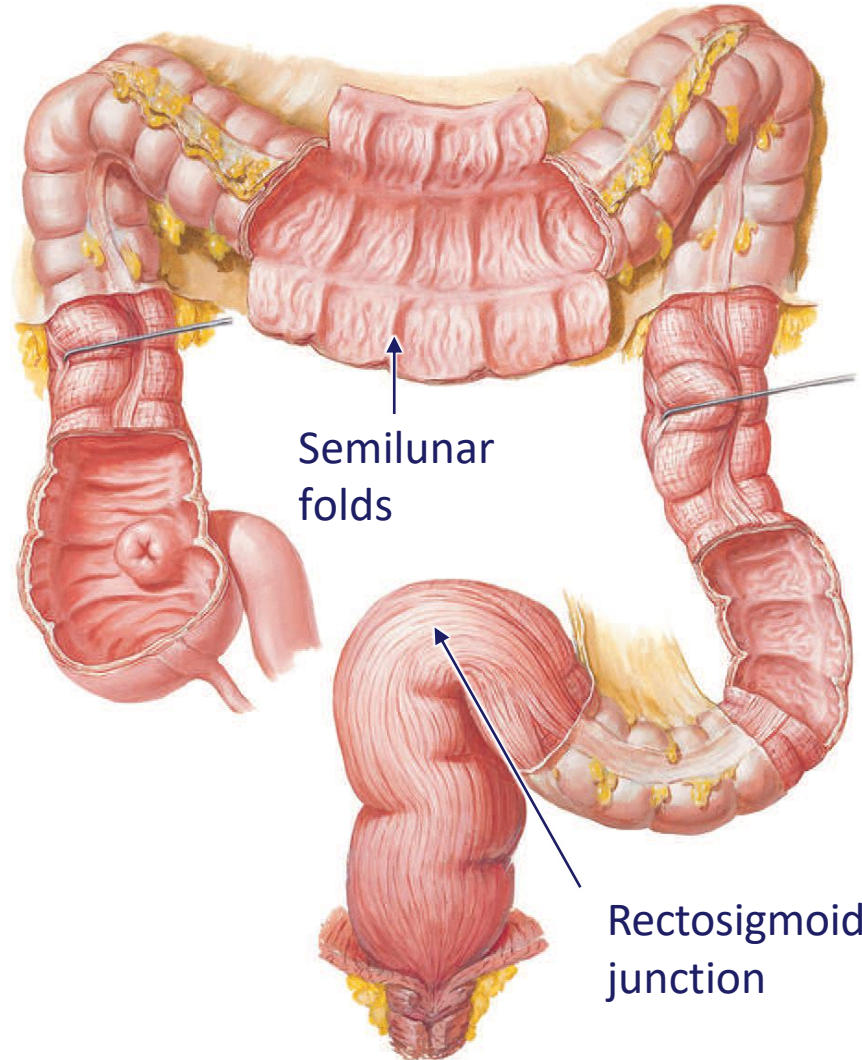


Appendicular artery in mesoappendix



Ileocecal valve

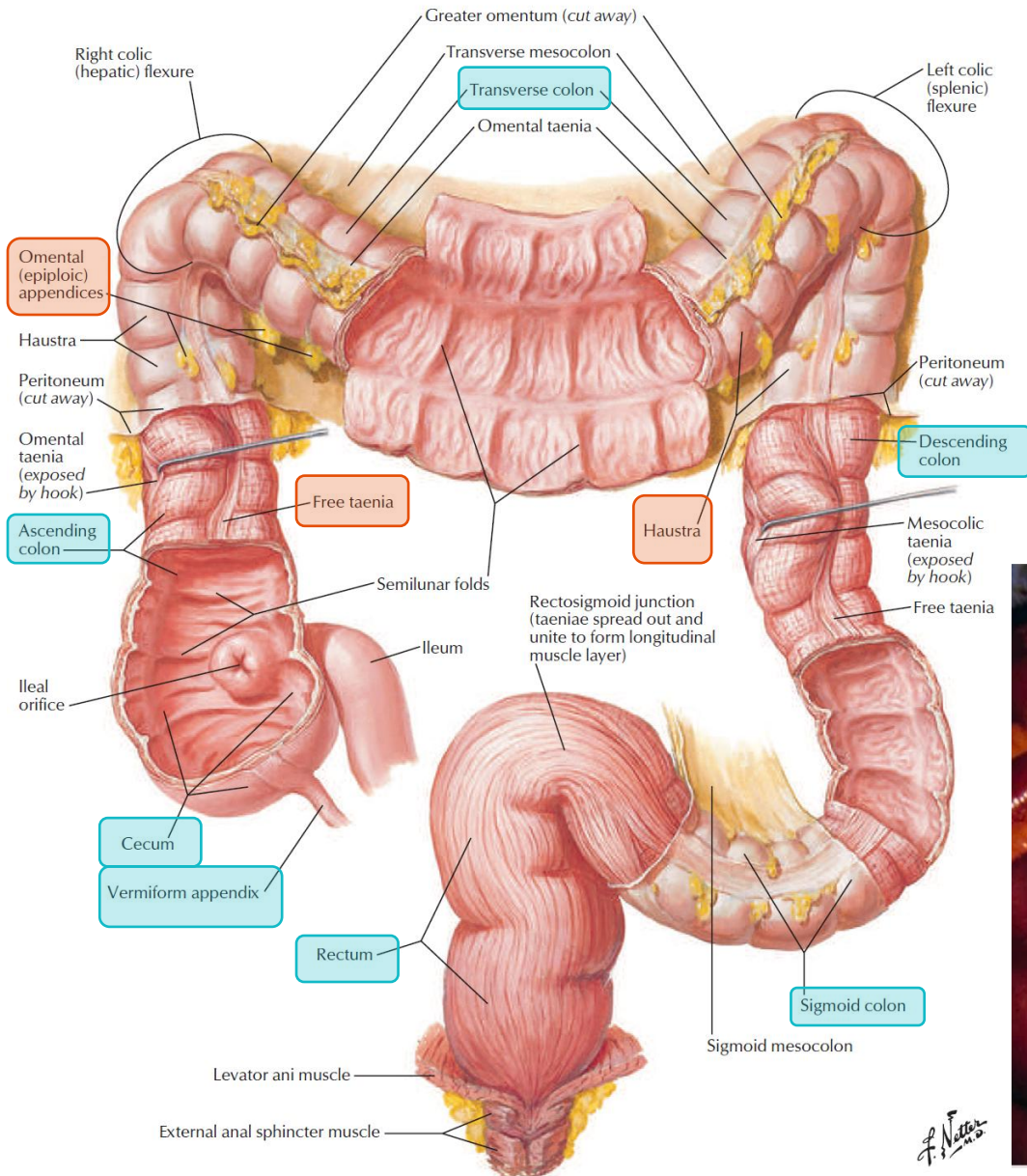
Appendix



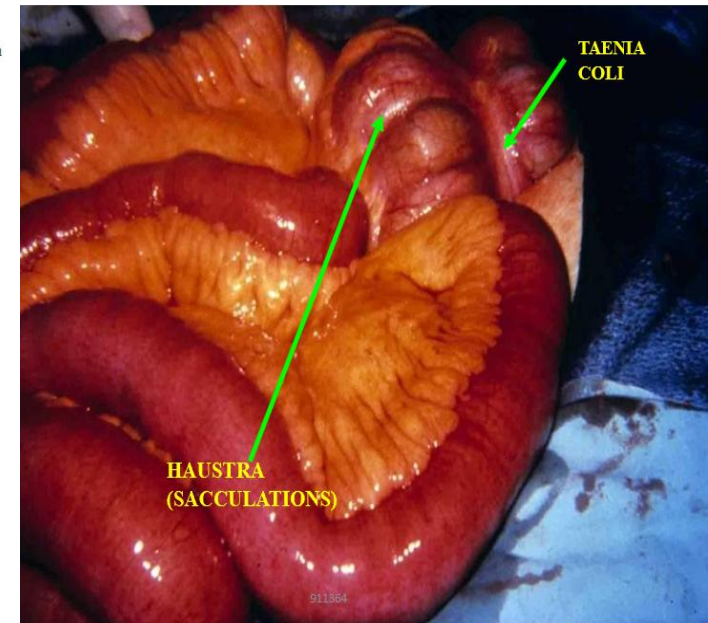
Semilunar folds

Rectosigmoid junction

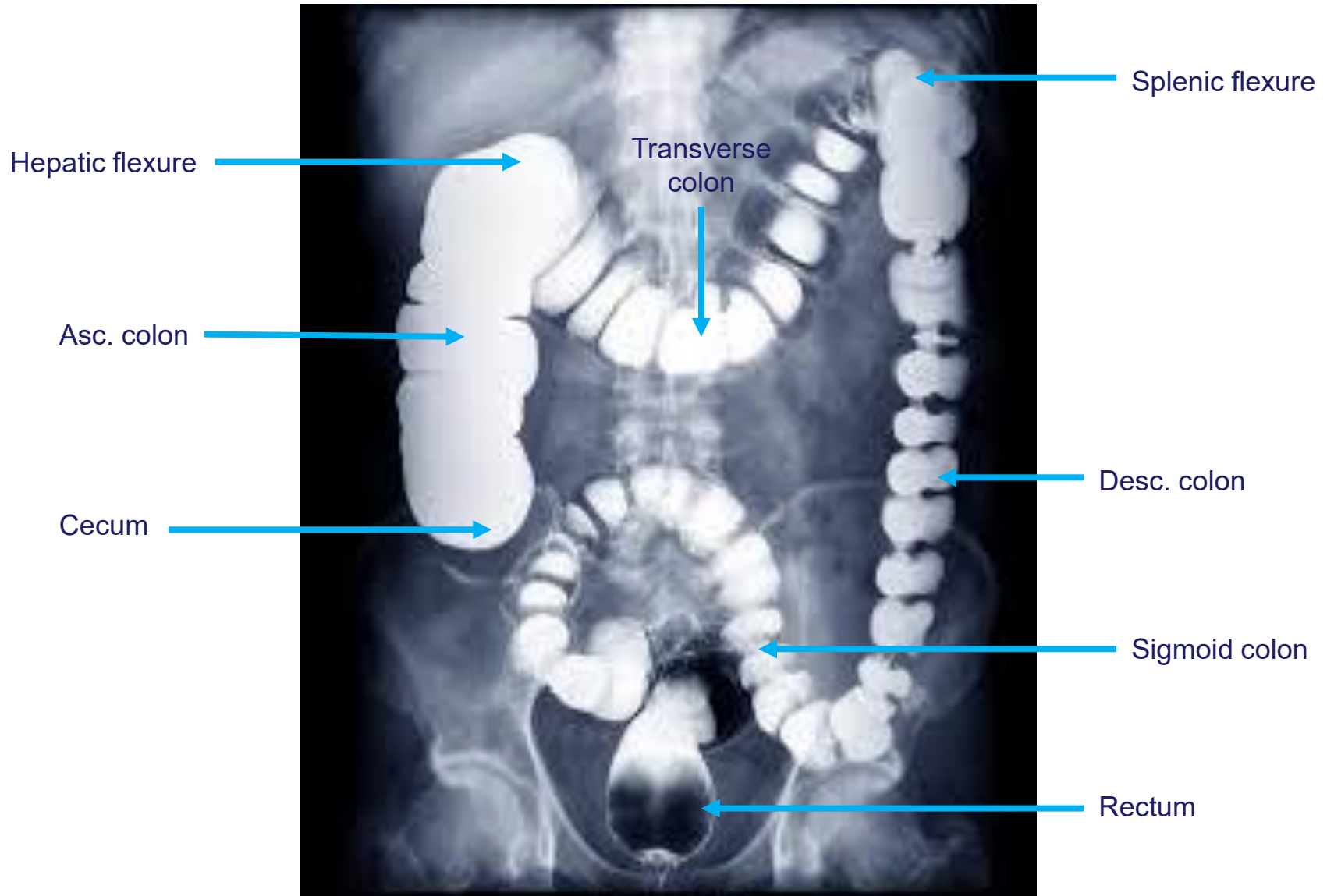
Large Intestine



- **Omental (epiploic) appendices**
 - Peritoneum-covered fat accumulations
- **Taeniae coli**
 - Aggregate of longitudinal muscle
- **Haustra**
 - Sacculation of colon



Large Intestine - Imaging



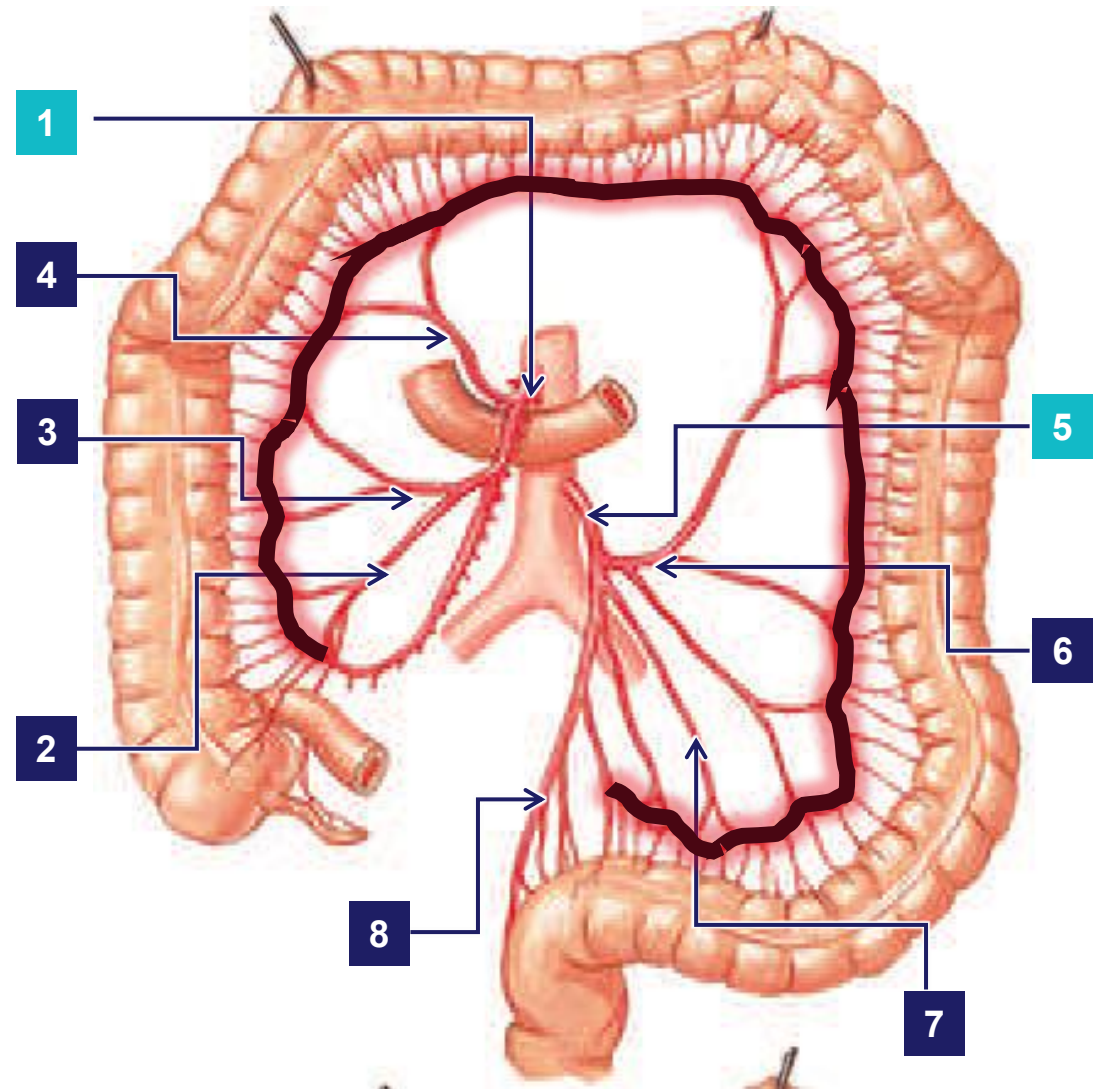
Large Intestine – Blood Supply

1. **Superior Mesenteric Artery**
2. Ileocolic
3. Right colic
4. Middle colic
5. **Inferior Mesenteric Artery**
6. Left colic
7. Sigmoidal
8. Superior rectal

Marginal artery

Watershed Area: a border zone supplied by arteries from distinct arterial sources

- 1) Splenic flexure (Griffiths point) branches of the SMA and IMA
- 2) Rectosigmoidal junction (Sudecks point)



Large Intestine – Watershed areas

Clinical significance

Vulnerability to Ischemia: While the marginal artery (of Drummond) connects these regions, they are the most susceptible to ischemic damage during SMA obstruction, heart failure, shock, or DIC (disseminated intravascular coagulation).

Ischemic Colitis: These watershed zones are classic locations for ischemic colitis, a condition where tissue death occurs due to insufficient blood supply.

The splenic flexure will likely experience ischemic injury due to blockage of the SMA

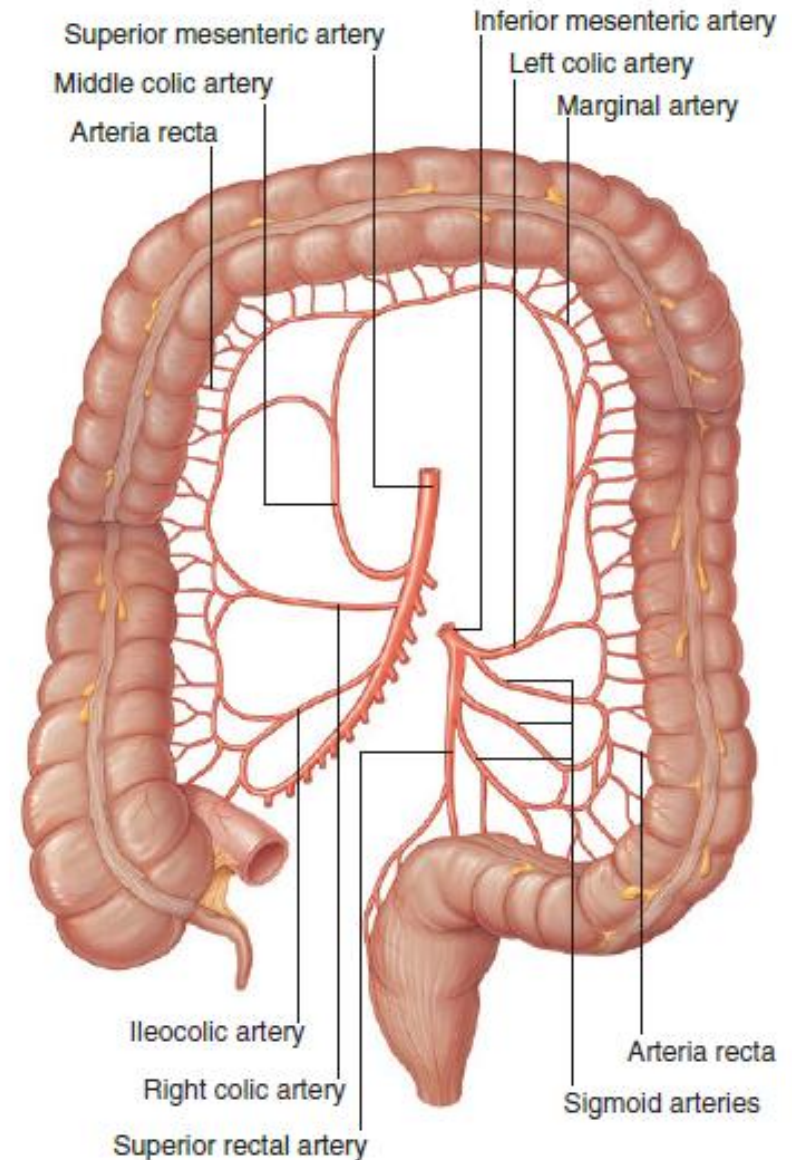
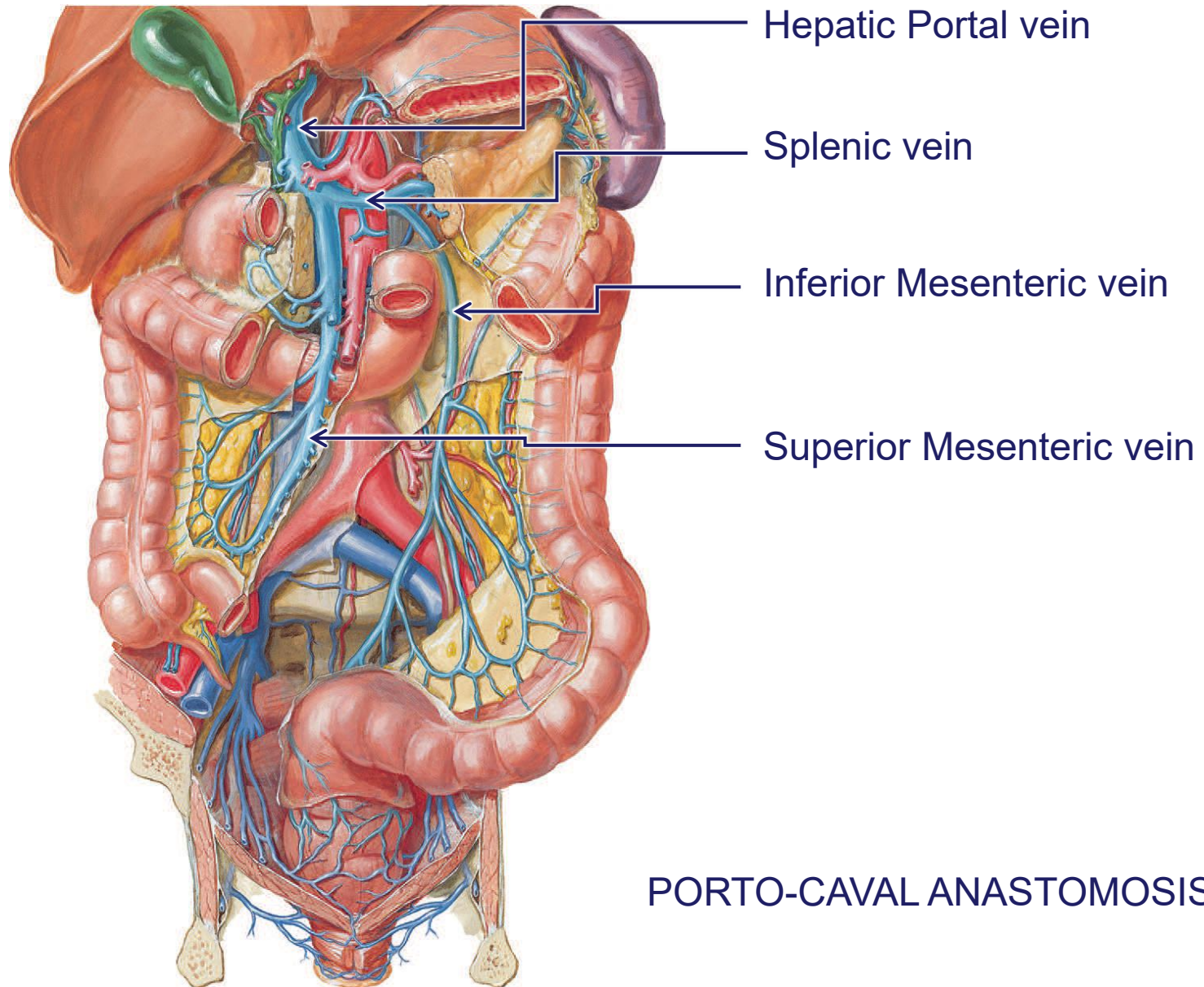


Fig. 4.90 Arterial supply to the colon.

Large Intestine – Venous Drainage



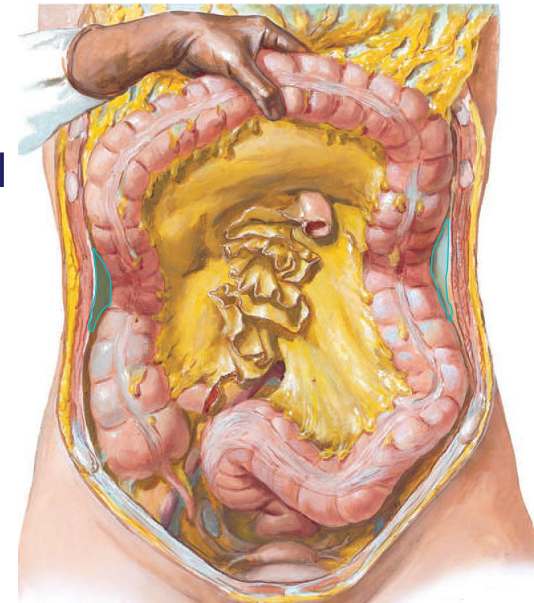
Surgical considerations

• Right + Left Paracolic gutters

- Depression in lateral margins of abdomen
- Between lateral margin of colon and posterolateral abdominal wall

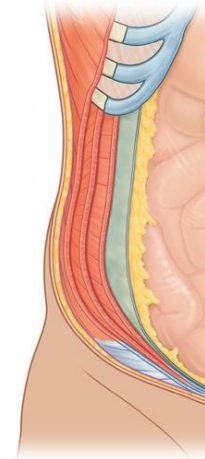
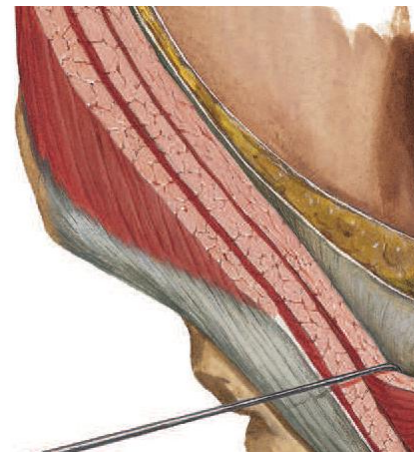
• Surgery: cut peritoneum along lateral paracolic gutter

- Devoid of major blood vessels and lymphatics
 - Relatively blood-free mobilization of ascending/descending colon



• Layers cut during surgery:

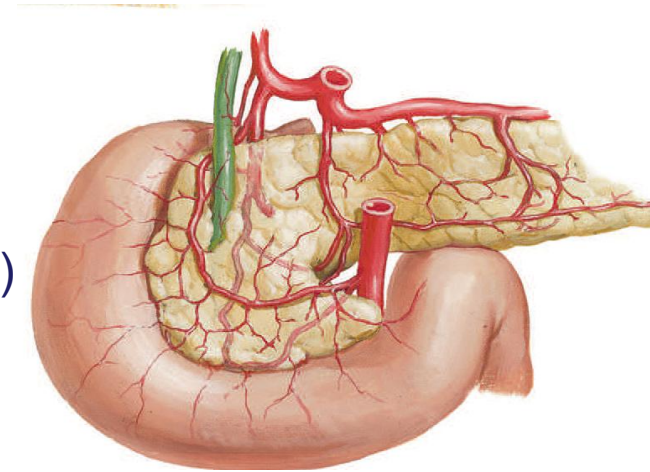
- External oblique muscle
- Internal oblique muscle
- Transversus abdominis m.
- Transversalis fascia
- Extraperitoneal fascia
- Parietal peritoneum



Arterial Anastomoses

Duodenum + Pancreas

- **Foregut**
 - Superior pancreaticoduodenal a. (Celiac trunk)
- **Midgut**
 - Inferior pancreaticoduodenal a. (SMA)

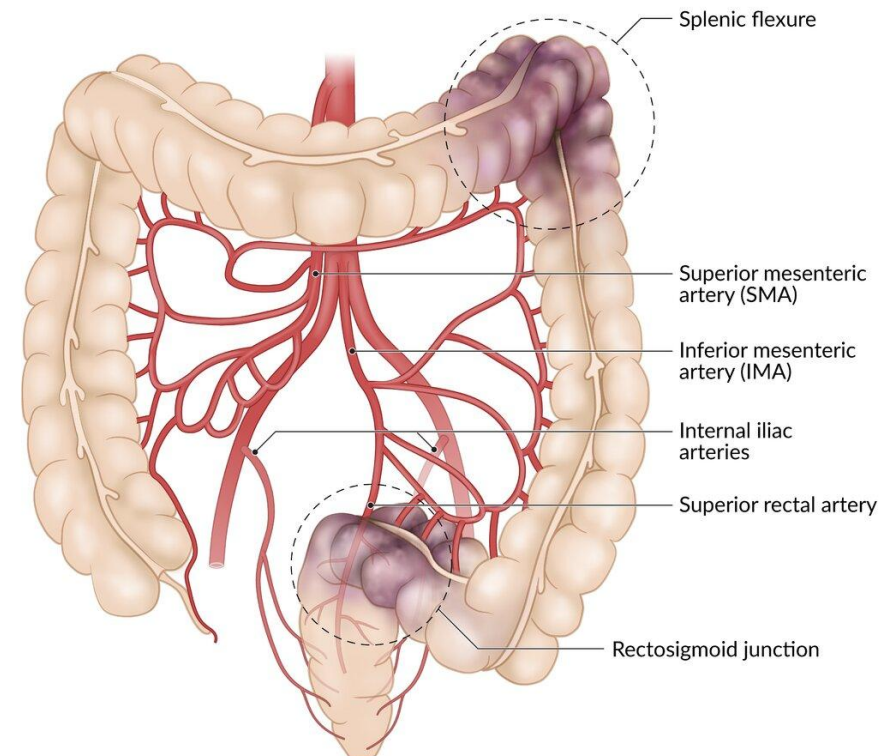


Transverse colon

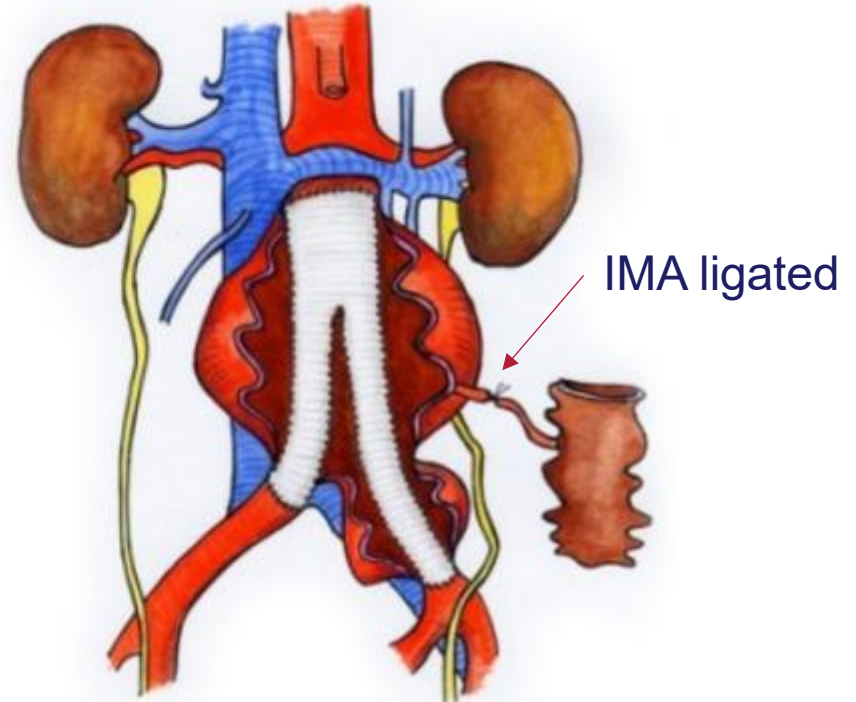
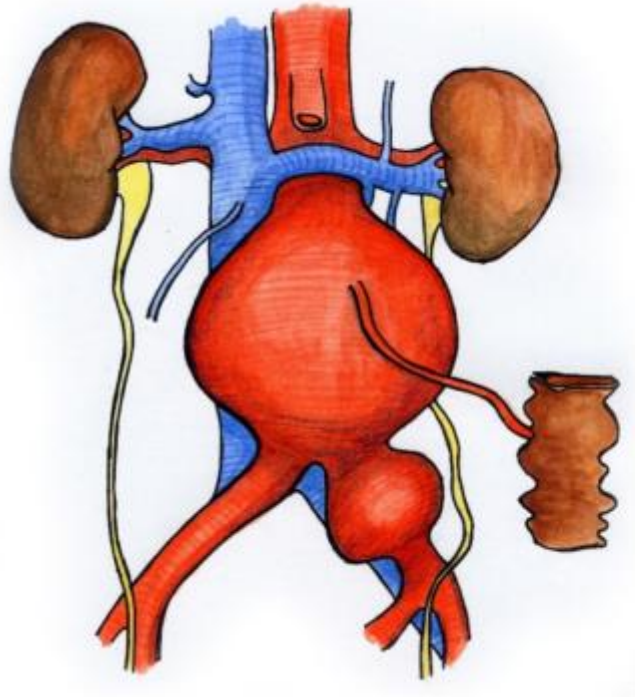
- **Midgut** - Middle colic (SMA)
- **Hindgut** - Left colic (IMA)
- **Marginal A. of Drummond**

Rectum

- **Hindgut** – Superior rectal (IMA)
- **Iliac** – Middle rectal (Internal Iliac)



AAA Repair



[195-23_Open-abdominal-aortic-aneurysm-AAA-repair.pdf \(uhd.nhs.uk\)](#)

Diverticulosis

- Outpouching of colonic mucosa and submucosa that herniates through muscular layer. Generally found in the **sigmoid colon**
- May erode into colonic vessels causing bleeding
- May be asymptomatic. If erosion into the colonic vessels takes place, painless rectal bleeding occurs.

Investigations:

- Labs: Anemia or positive stool guaiac test during bleeding.
- Radiology: diverticular seen on barium enema and colonoscopy.

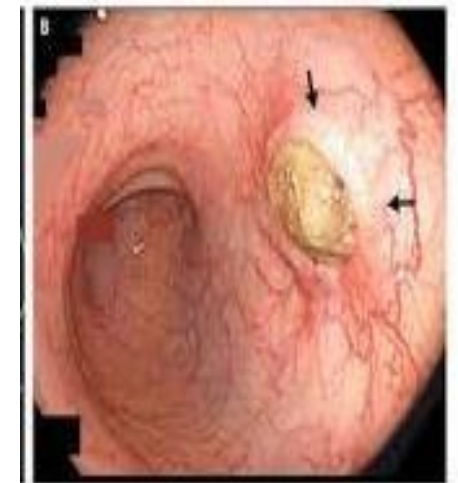
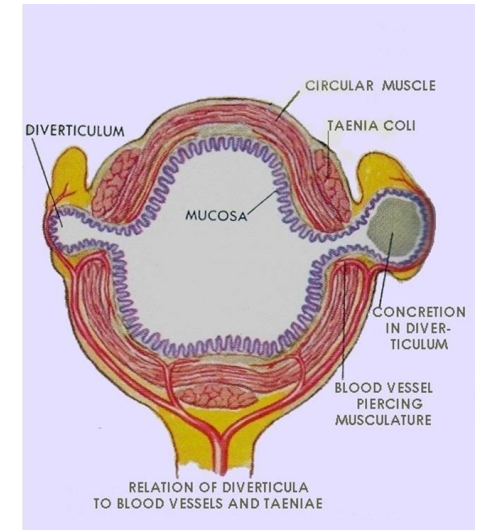
Complications:

- **Diverticulitis**: Obstruction of the diverticulum leading to infection.
- **Perforation**: rupture may lead to peritonitis, fistula formation, pelvic abscess

Clinical picture:

- Fever
- Left lower quadrant pain
- Abdominal distention; Abdominal tenderness

Terms to remember: melena, hematochezia and meconium



Large Bowel Obstruction

- Tends to be located on the **periphery**.
- Increased intraluminal gas
- Presence of prominent haustra coli
- Obstruction at any point along the gut tube would cause the part of the tube proximal to the point of the obstruction to be dilated and the part located distally to be flattened.

Clinical Features.

- Nausea and vomiting
- Abdominal distention
- Increase bowel sounds
- No passage of gas per rectum (obstipation)



Sigmoid Volvulus



- Twisting of the sigmoid colon around its mesentery (the sigmoid mesocolon), causing obstruction of the colon.

Clinical Features:

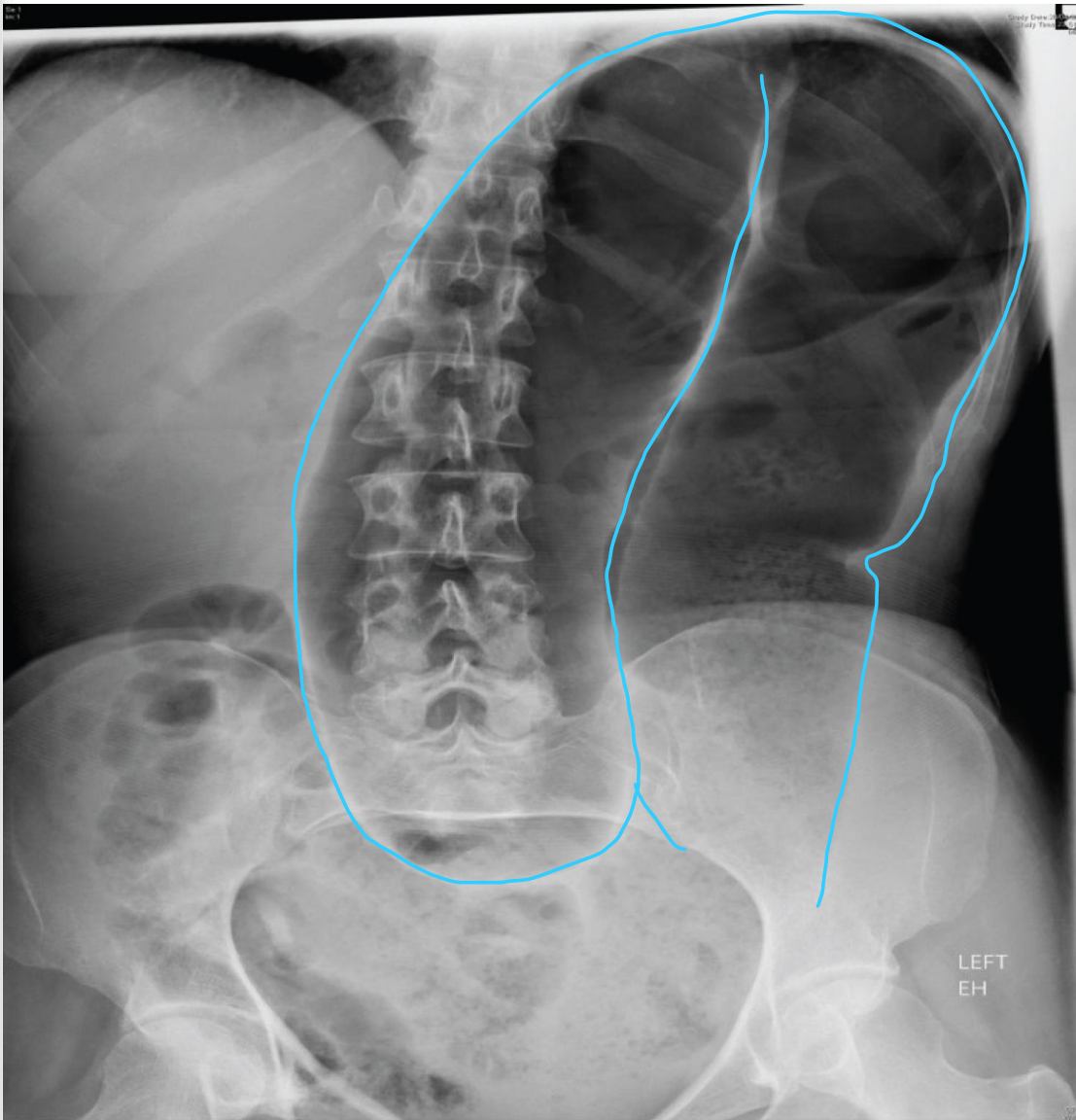
- Abdominal pain
- Nausea and vomiting
- History of constipation

Complications:

- ischemia of colon
- Perforation of colon
- Peritonitis

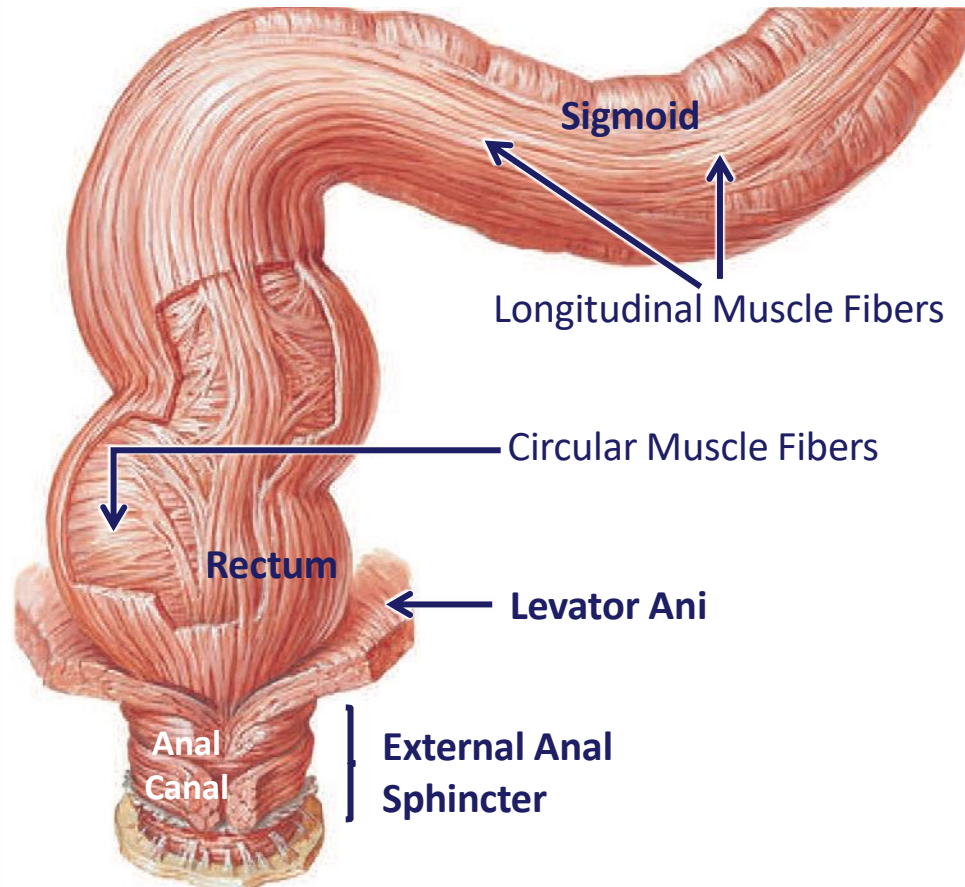
N.B. can also occur in small bowel as it has a mesentery

Sigmoid Volvulus – Imaging Plain Film



- Double loop obstruction.
- “Coffee bean” sign
- May have symptoms of both large and small bowel obstruction depending on severity of the obstruction.
- No gas seen in the rectum

Rectum



- Continuation of the GI tract into the **pelvis**
- Begins at the rectosigmoidal junction
- Terminates at the anorectal junction
- The rectum has no mesentery
- Superior 2/3 is retroperitoneal and forms the posterior surface of the rectovesical and rectouterine pouches
- Inferior 1/3 is sub peritoneal

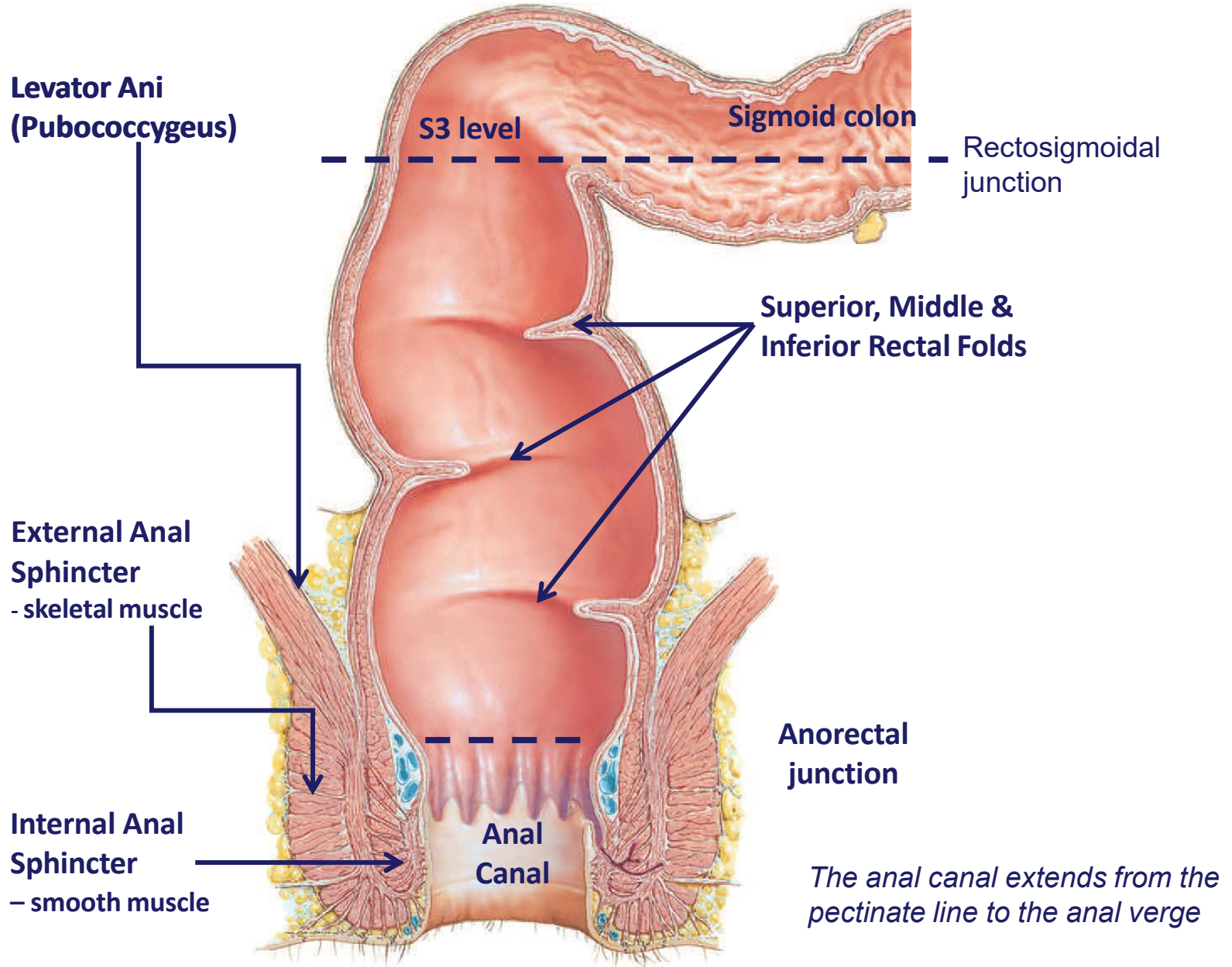
Internal structure:

- 3 transverse folds
- **2 on the left and 1 on the right**

Ampulla

- Most distal part of the rectum stores feces until defecation
- Puborectalis forms a sling around the **anorectal junction**

Rectum & Anal Canal



Clinical correlate: Rectocele

Rectocele: Herniation of the rectal wall, with the most common type being an anterior rectocele where rectum protrudes into the posterior vaginal wall due to defects in the recto-vaginal septum.

Risk factors:

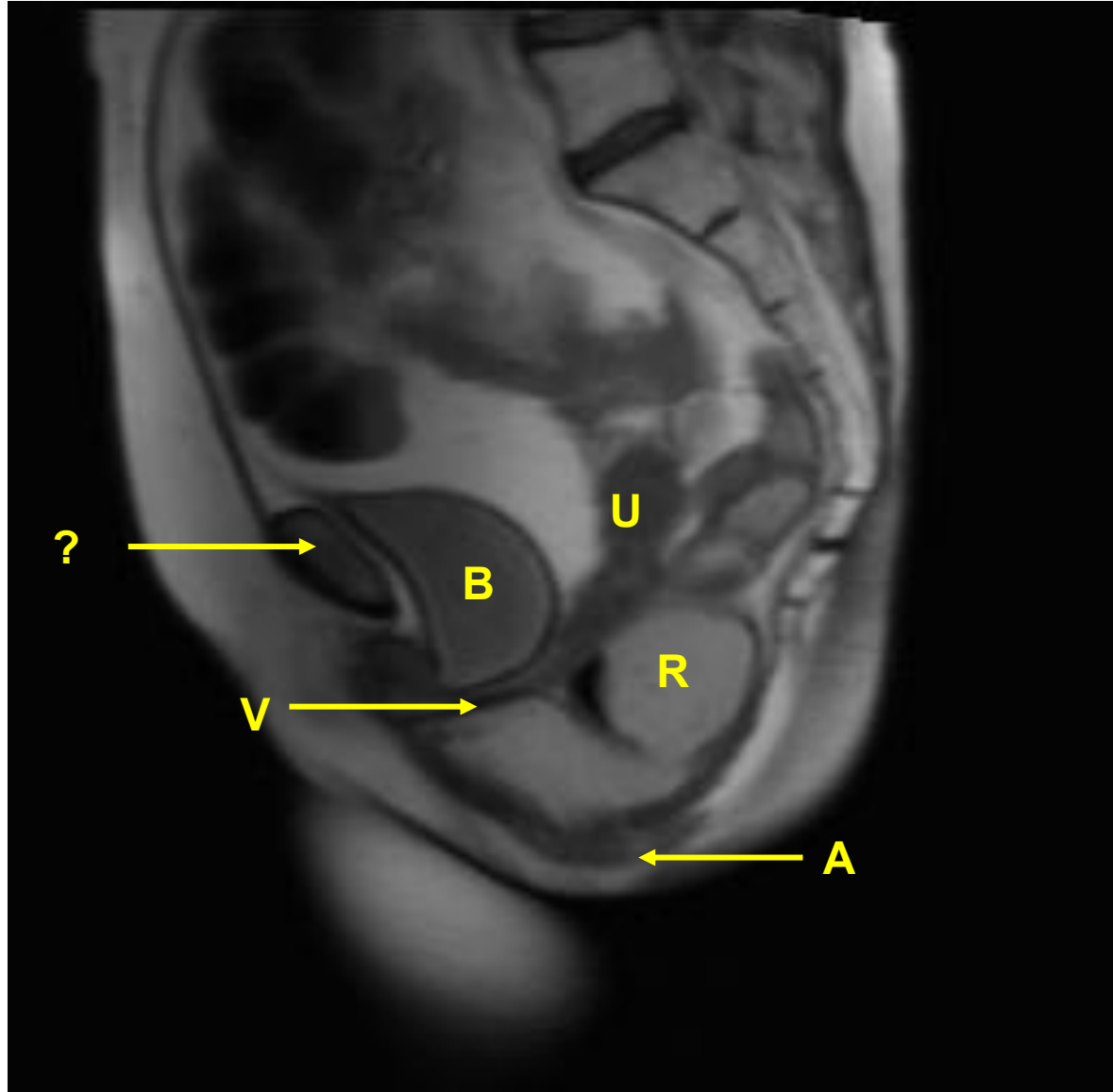
- obstetric trauma
- chronic constipation
- Smoking

Presentation

- difficult defecation / excessive straining
- vaginal pressure and bulging
- intermittent fecal soiling
- necessity to digitate to allow the evacuation of stools



Clinical correlate: Rectocele



Pectinate line

The pectinate line (also known as the dentate line) is **a line which divides the upper two-thirds and lower third of the anal canal**

- **Appearance:** Often appears as a whitish line, sometimes called *Hilton's white line*.
- **Location:** About halfway up the anal canal, where the anal valves meet
- **Developmental Junction:** Marks the boundary where the primitive hindgut joins the proctodeum (an embryonic structure).
- **Tissue Change:** Separates the upper canal (columnar epithelium) from the lower canal (stratified squamous epithelium).
- **Clinical Importance:**
 - **Above the line:** Visceral innervation (dull diffuse poorly localized), internal iliac lymph nodes drainage.
 - **Below the line:** Somatic innervation (sharp, intense well localized), superficial inguinal lymph nodes drainage, external hemorrhoids.

Pectinate Line

Levator ani (striated muscle)

nn. of sacral plexus

Internal anal sphincter (smooth muscle)

SNS – constriction PSNS – relaxation

External anal sphincter (striated muscle)

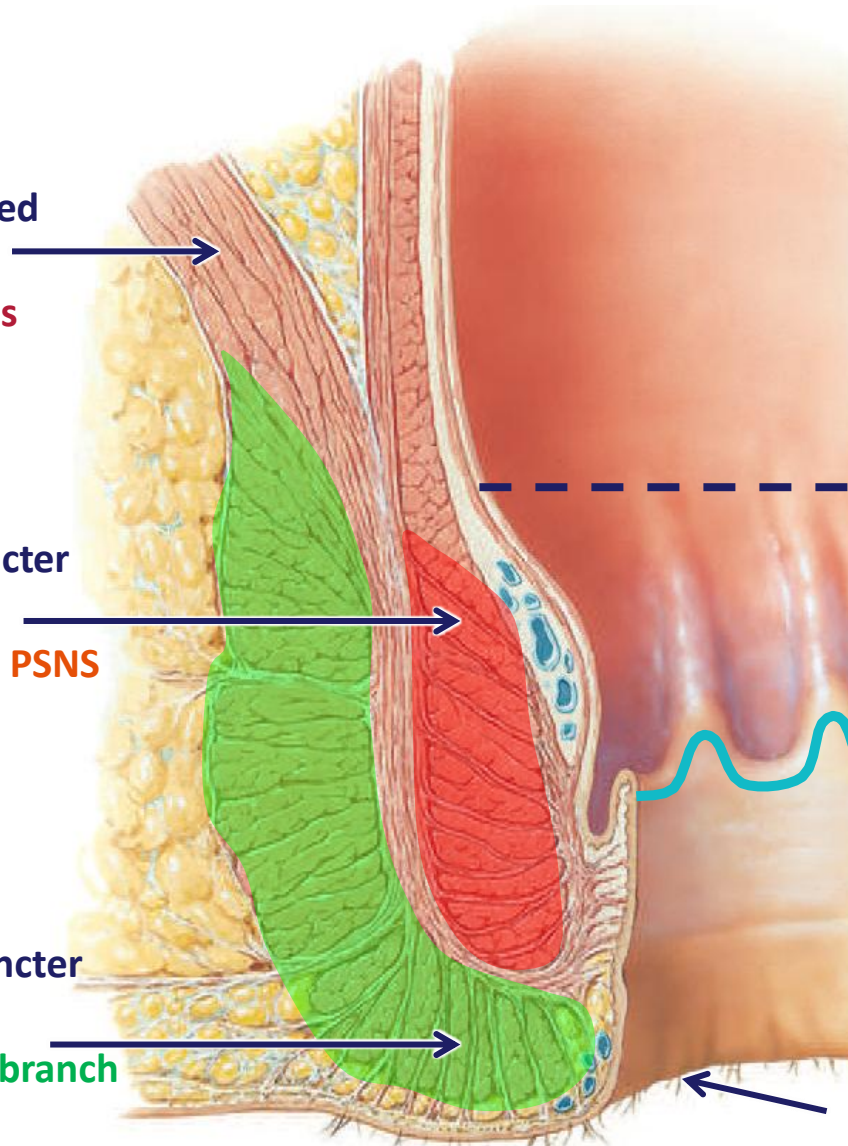
Inferior rectal n. (branch of pudendal n.)

Endodermal origin – visceral afferents

Pectinate Line

Ectodermal origin – somatic afferents

Anal verge



Pectinate Line

Levator ani
(striated muscle)
nn. of sacral plexus

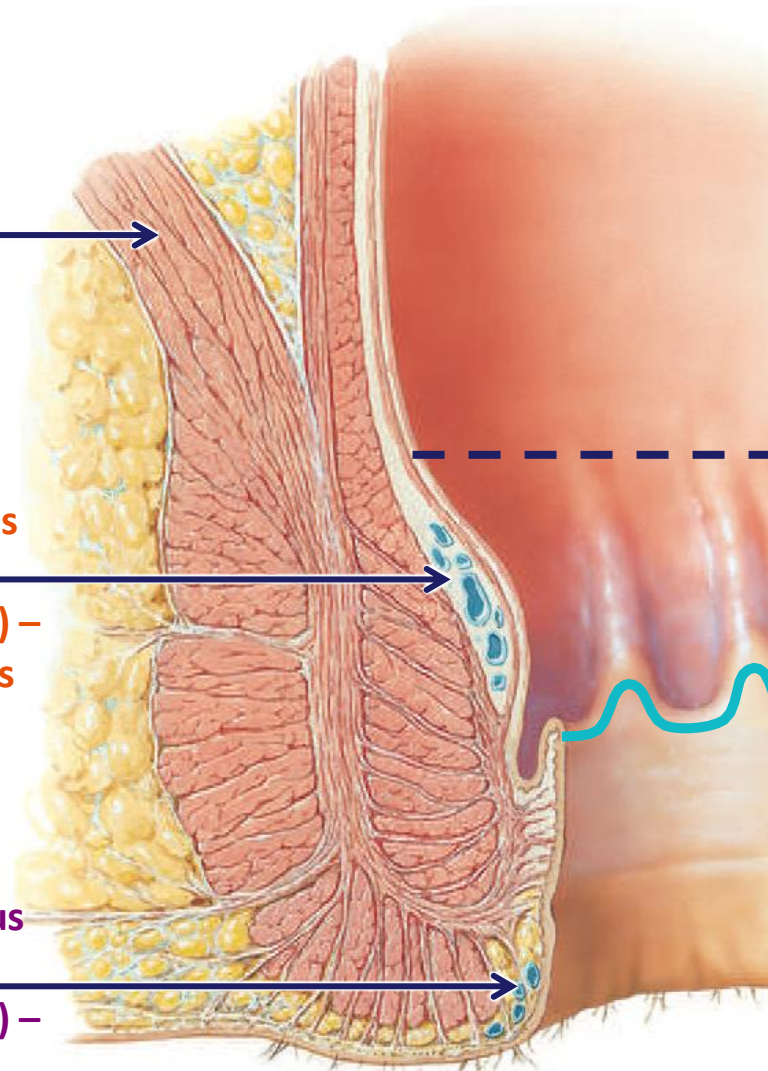
Internal rectal venous
plexus ("internal
hemorrhoidal veins") –
painless hemorrhoids

External rectal venous
plexus ("external
hemorrhoidal veins") –
painful hemorrhoids

Endodermal
origin – visceral
afferents

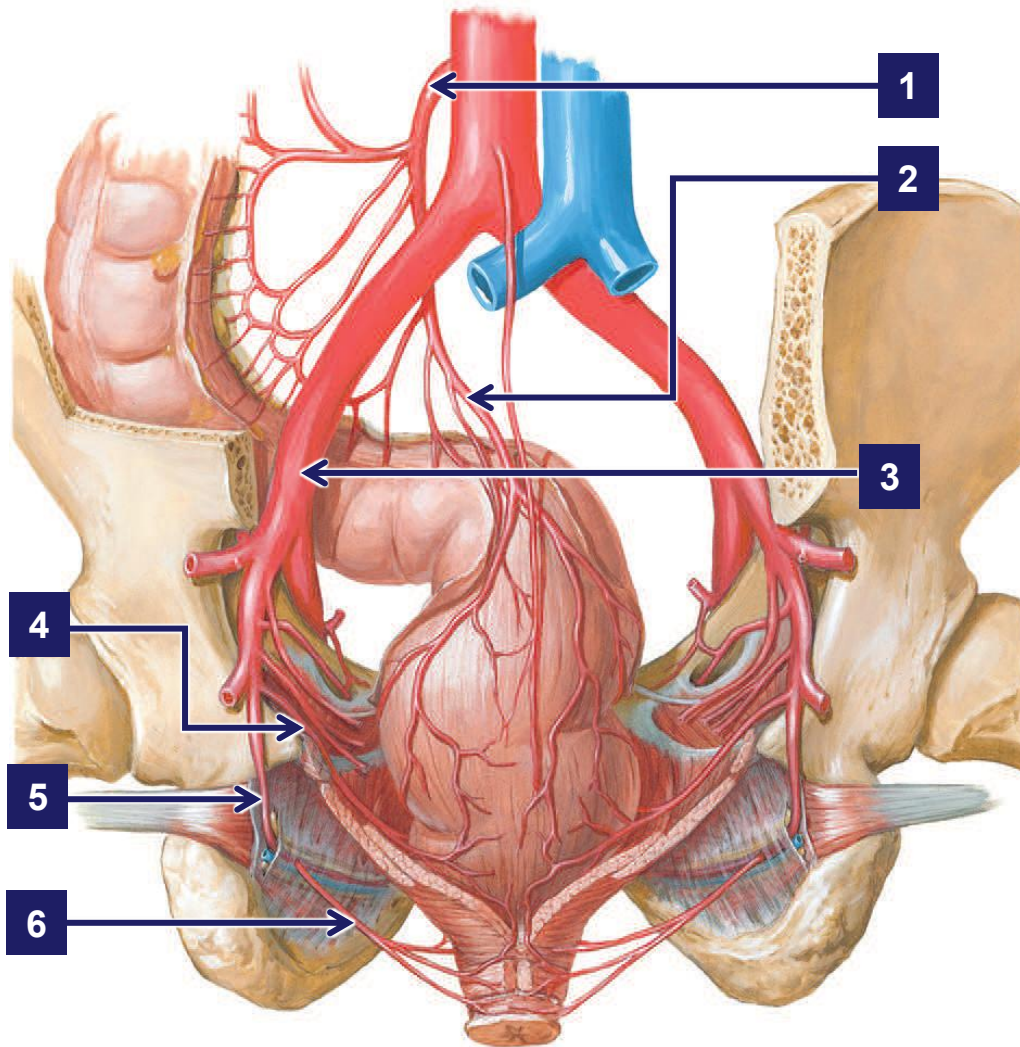
Pectinate Line

Ectodermal
origin – somatic
afferents



- **Summary:** The anal canal above the pectinate line differs in its embryonic origin, arterial supply, innervation, venous and lymphatic drainage.

Arterial Supply of Rectum



1 – Inferior mesenteric; 2 – Superior rectal; 3 – Internal iliac; 4 – Middle rectal; 5 – Internal pudendal; 6 – Inferior rectal

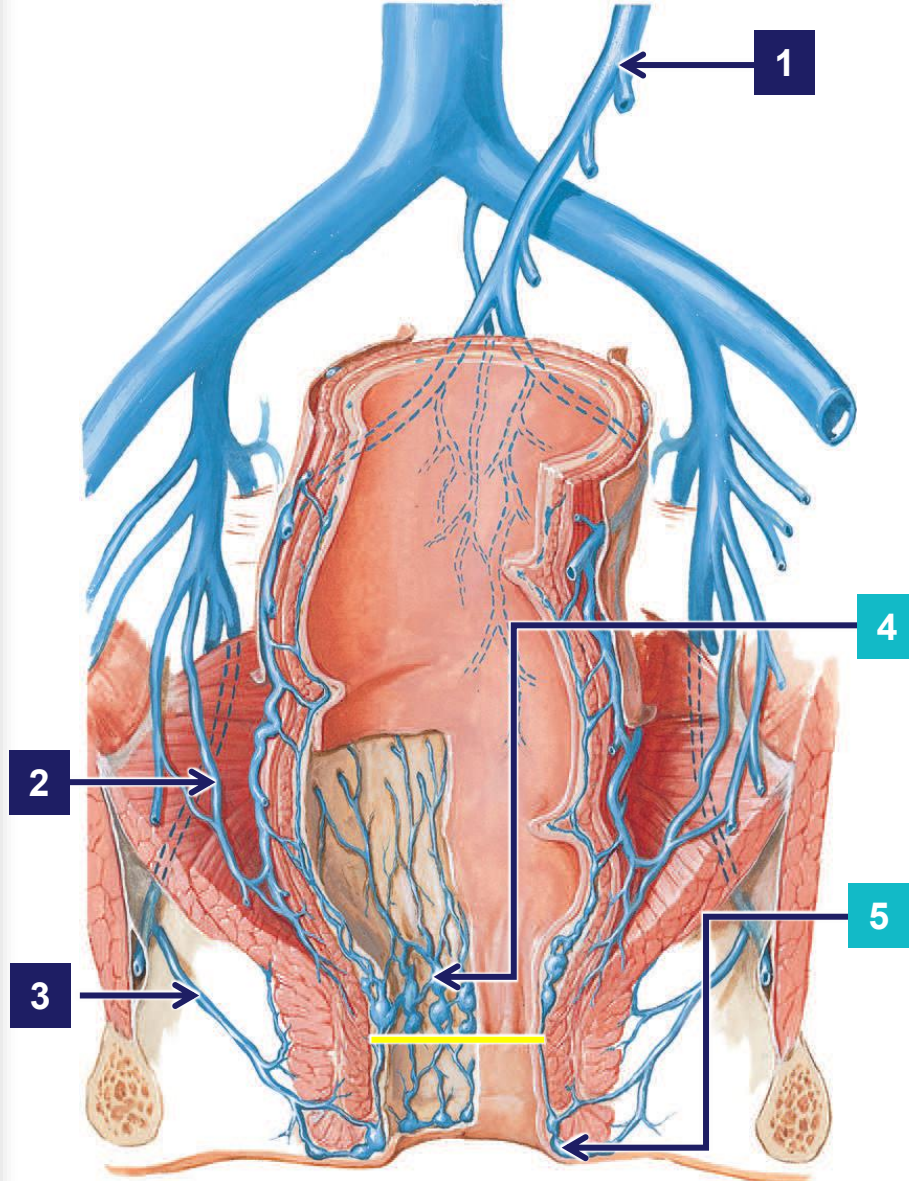
Rectum & anal canal are supplied by superior, middle and inferior rectal arteries

- **Superior rectal artery:** branch of the Inferior mesenteric a.
- **Middle rectal artery:** branch of the internal iliac a.
- **Inferior rectal artery:** branch of the internal pudendal a.

Therefore, around the rectum there is a rich anastomosis due to branches of:

- Inferior mesenteric a.
- Internal iliac a.
- Internal pudendal a.

Venous Drainage of Rectum



1 – Superior rectal ; 2 – Middle rectal; 3 – Inferior rectal

- The internal rectal plexus drains in both directions at the level of the pectinate line [yellow line]
- Superior to the pectinate line the **internal** venous plexus drains into the superior rectal vein (inferior mesenteric → portal vein)
- Inferior to the pectinate line the **external** venous plexus drains into the inferior rectal veins (caval system i.e., IVC)

Internal hemorrhoids [4]:

- Prolapses of the rectal mucosa containing dilated veins of the internal rectal plexus.

External hemorrhoids [5]:

- Dilations of the external rectal plexus.

Predisposing factors for hemorrhoids:

- Any disorder that impedes venous return - Increase in intra-abdominal pressure

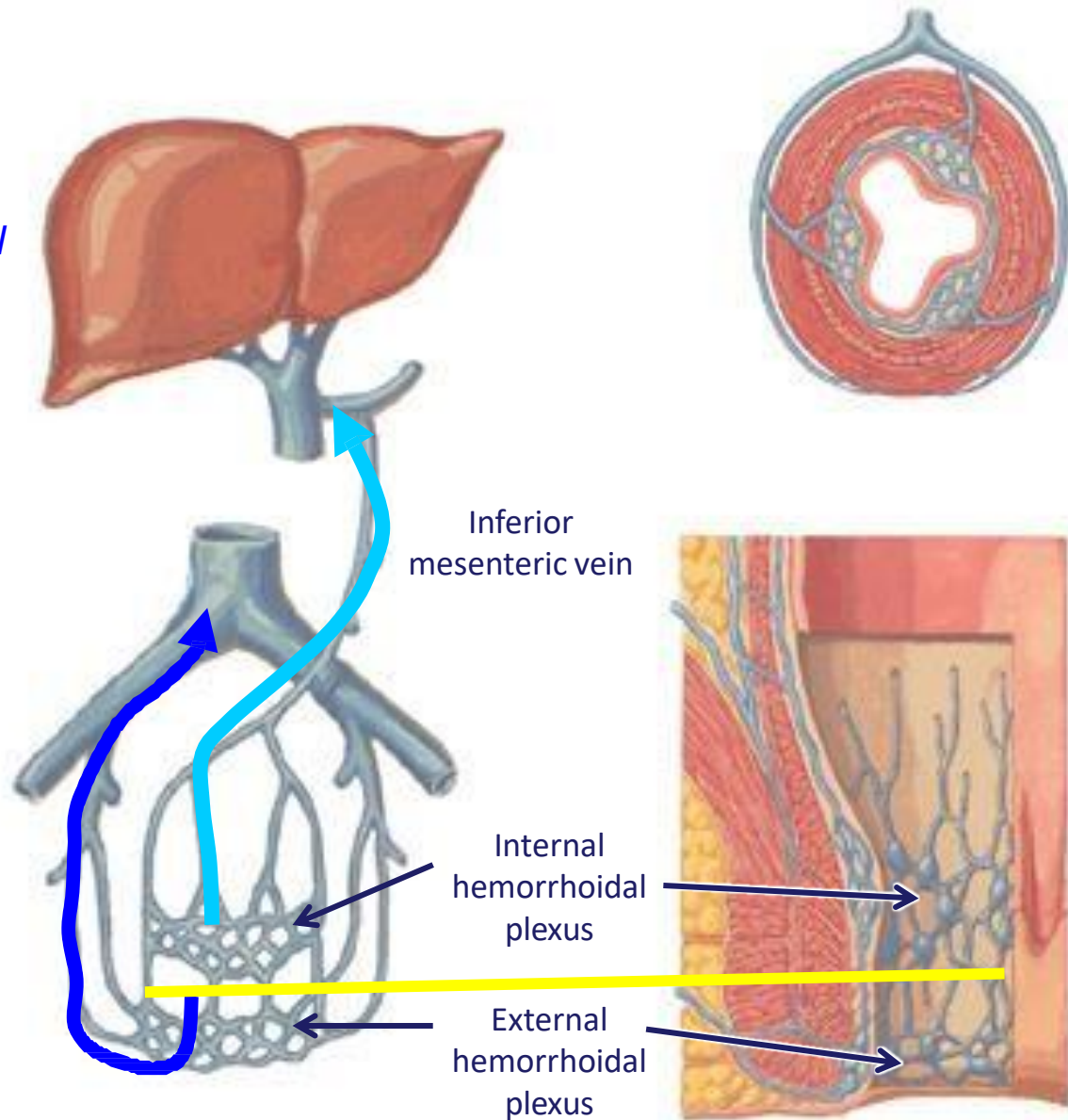
Clinical considerations (Rectum)

Lymphatics

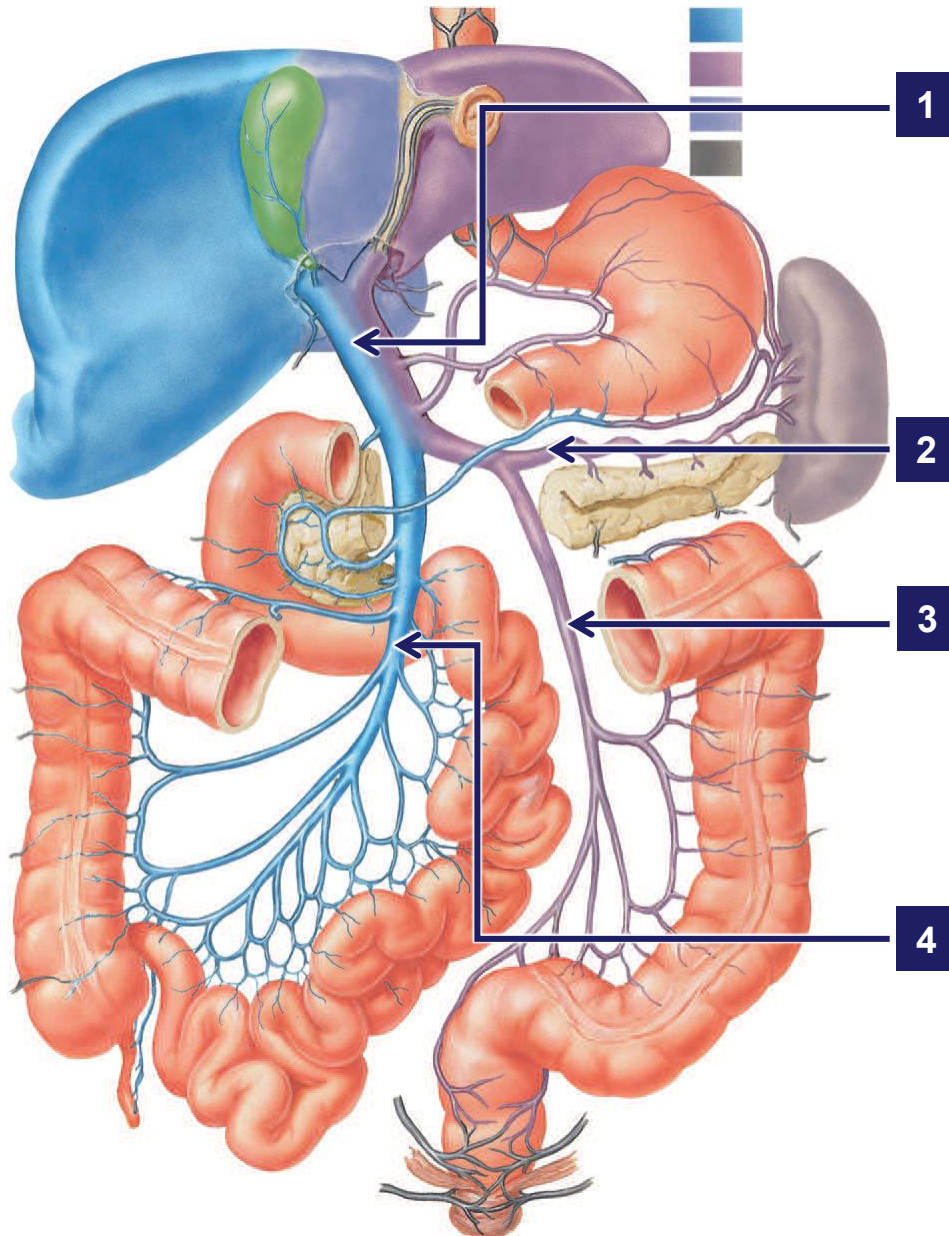
- **Above** the pectinate line lymph drains to the *internal iliac nodes*.
- **Below** the pectinate line lymph drains to the *superficial horizontal inguinal lymph nodes*

Clinical correlate:

- **Rectal route for drugs** – (Rectal suppositories)
- **Portal Hypertension:** In this condition, the portocaval anastomoses between the superior (portal), middle and inferior rectal (caval) veins become varicosed.
- **Hematochezia** – the passage of fresh blood per anus, usually mixed with or on stool indicative of a lower GI bleed.



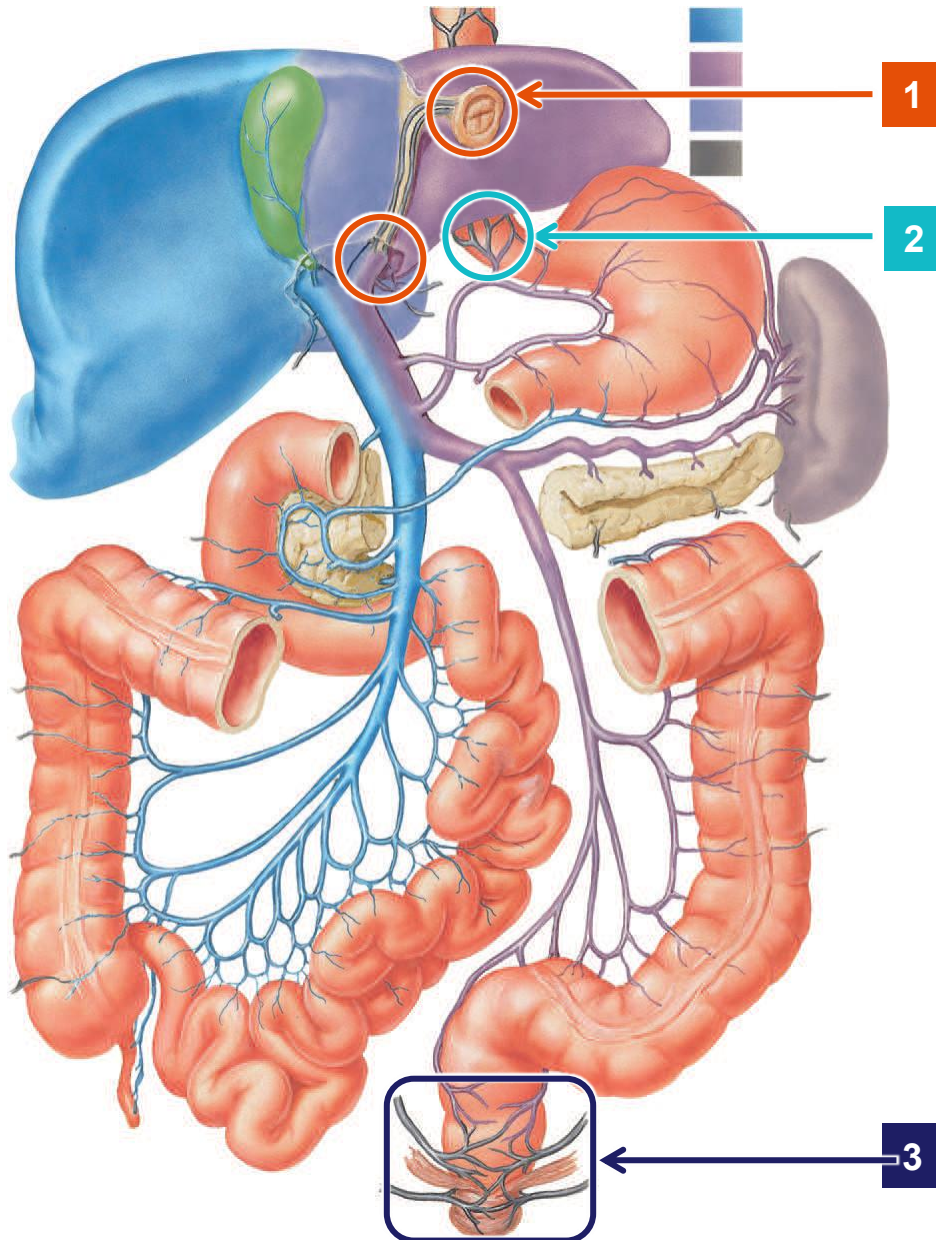
Hepatic Portal Vein



- Collects venous blood with products of digestion from the abdominal part of GIT + gallbladder, spleen & pancreas
- Passes through the lesser omentum, enters the liver through porta hepatis, divides into right & left and then segmental branches → ends in venous sinusoids of liver
- Formation – behind the neck of pancreas; joining of splenic vein & superior mesenteric vein

1 – Hepatic Portal vein; 2 – Splenic vein; 3 – Inferior mesenteric vein; 4 – Superior mesenteric vein

Portocaval Anastomoses



1. **Paraumbilical (Caput Medusae)**
 - Paraumbilical veins (portal)
 - Superficial epigastric veins (caval)
2. **Lower Esophagus (Esophageal varices)**
 - Esophageal tributaries of Left Gastric vein (portal)
 - Esophageal tributaries of azygos + hemiazygos veins (caval)
3. **Rectal (Hemorrhoids)**
 - Superior rectal veins (portal)
 - Middle + Inferior rectal veins (caval)
4. Retroperitoneal
5. Bare area of liver

Portocaval anastomosis summary table

Name of clinical condition	Location	Portal circulation	Systemic circulation
Esophageal varices	Lower esophagus	Esophageal branch of left gastric vein	Esophageal branches of azygos vein
Rectal varices/Hemorrhoids	Rectum	Superior rectal vein	Middle rectal veins and inferior rectal veins
Caput medusae	Paraumbilical	Paraumbilical veins	Superficial epigastric vein
Splenorenal shunt	Retroperitoneal	Splenic vein	Renal vein, suprarenal vein, paravertebral vein, and gonadal vein
No clinical name		Right colic vein, middle colic vein, left colic vein	Retroperitoneal veins of Retzius
No Clinical name	Bare area of liver posterior right lobe of liver	Small portal vein tributaries	Inferior phrenic veins

Lymphatic Drainage

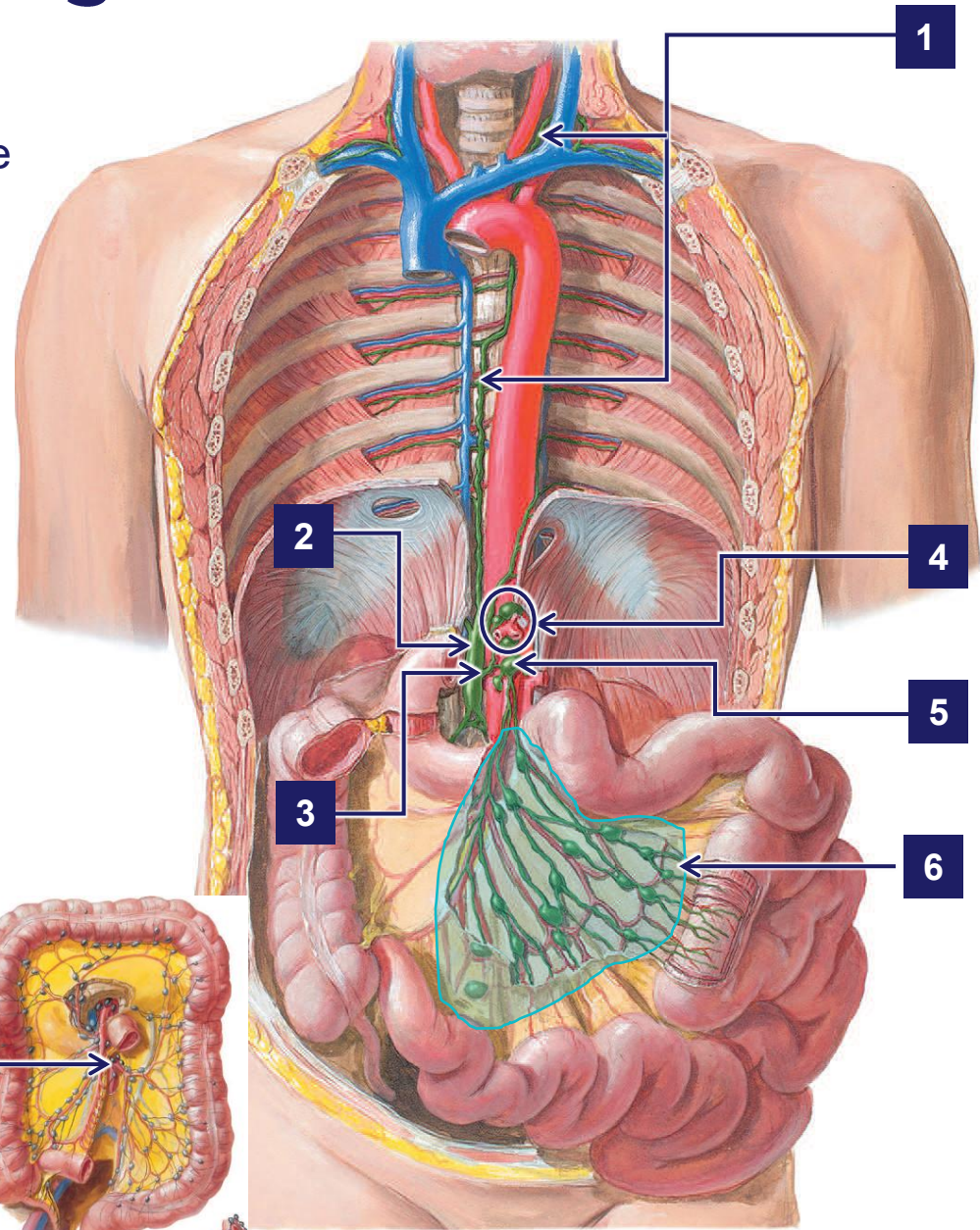
Thoracic Duct

- Drains **lymph** of **lower half** and **upper left quadrant** of the body and **ALL** of the viscera via **pre-aortic lymph nodes**

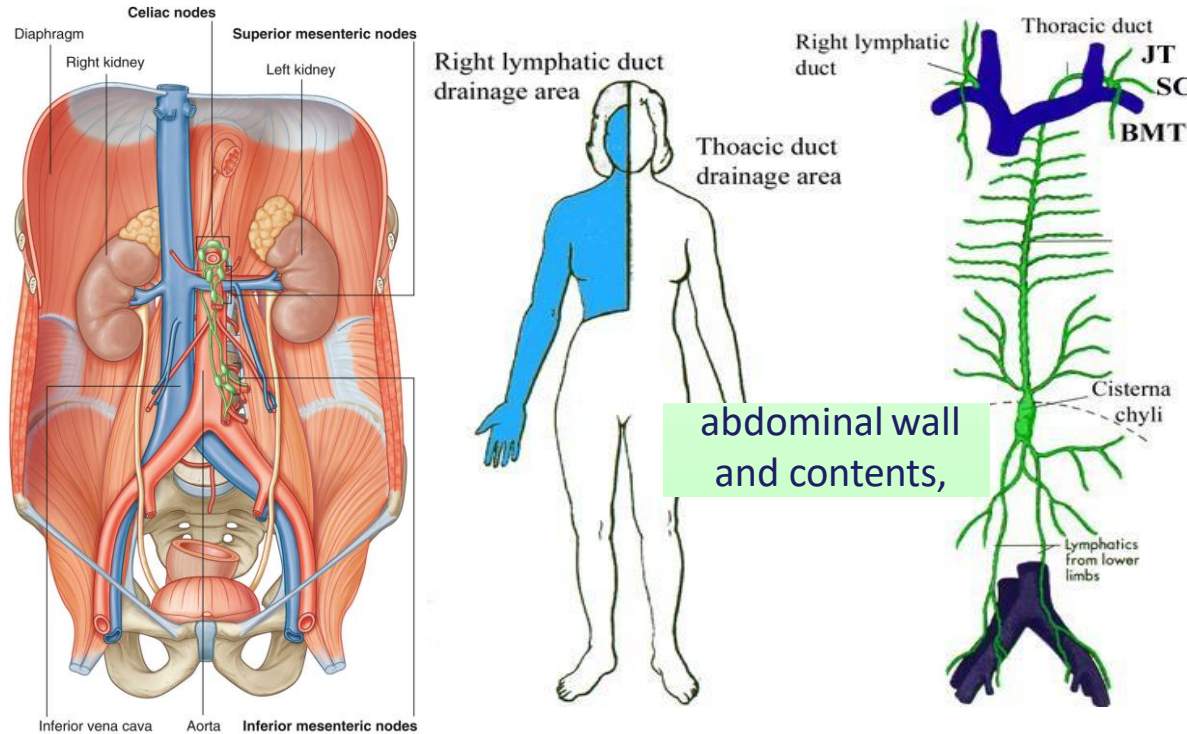
Cisterna Chyli

- Sacculation at the lower end (beginning) of thoracic duct.

- Thoracic Duct (empties into left jugulo-subclavian junction)
- Cisterna Chyli
- Intestinal trunk
- Celiac nodes**
Superior mesenteric nodes
- Central group
- Juxtaintestinal group
- Inferior mesenteric nodes**



Lymphatic Drainage



Right Lymphatic Duct

- Drains lymph of upper right quadrant of the body

Left supraclavicular node (Virchow's node): can be enlarged on physical examination due to distant metastasis from abdominal organs